

Data-driven careers

Adrian: Create a data-driven business solution

- A customer engineer at Google is the bridge between the technical and business components or aspects of corporate America

Data-driven careers drive modern business

Data professionals are so valuable to their companies. They determine which data streams are most important to specific business projects, challenges, and initiatives. They identified key goals for the future. They give their organizations the ability to take meaningful action by re-imagining processes and improving operations.

Two categories of data careers

- Technical
- Strategic

Technical data professionals

- Expertise in mathematics, statistics, and computing
- Build models and make predictions
- Explore datasets
- Technical data professionals are expert data analysts, machine learning engineers, and statisticians.

Now let's consider data professionals on the more strategic side. These people include business intelligence professionals and technical project managers, to name a couple.

Strategic data professionals

- Interpret information for organization's operations, finance, research, and development
- Work aligns with business strategy

Let's consider some examples of how data professionals use their expertise to transform industries such as finance, health care, manufacturing, and agriculture.

Finance

- Assess risks
- Monitor market trends
- Reduce fraud
- Create a more stable financial system

Healthcare

- Process clinical data
- Support early detection
- More precise diagnoses

Manufacturing

- Predict when to perform preventative maintenance
- Maximize quality assurance
- Respond to logistical issues
- Enable clear and near real time communication with suppliers, retailers, and other network partners

Agricultural

- Develop new approaches
- Improve harvesting technologies

Profiles of data professionals

You have been learning about data professionals and how valuable their work is to the companies that employ them. You also learned about technical and

strategic roles for data analytics professionals. **Data analysts in technical roles transform raw data into something useful for decision-making.** **Strategic data analytics professionals** focus on **maximizing information** to guide the businesses they are working for. In this reading, you will learn more about the data professional career space, and see examples of some general categories of data professional roles.

The classifications of data professional roles presented here are a very general introduction to the careers available within the data career space. When you begin to look for a job, you may encounter entry-level, mid and senior positions within each of these general categories.

Data scientist and data analyst

Data scientist and data analyst are roles that work directly with data. These professionals gather, clean, analyze, and share insights from data with stakeholders. An increasing number of industries turn to data analysis to create insights that inform various tasks like guide decision-making, identify user preferences, or determine how to use resources more effectively. With more industries looking for data professionals, there is a great chance that you will find one that matches your interests.

Key attributes

- **What they do:** Uncover trends, patterns, and insights from data
- **How they do it:** Employ advanced modeling and statistical analytics techniques
- Entry-level data scientist or analyst positions may not require considerable data analysis experience and be less restrictive about requiring those hired to hold a traditional degree.

Sample job titles

- Data scientist
- Marketing analyst
- Data analyst
- AI analyst
- Business analyst

Data management and infrastructure

Data professionals that work in data management and infrastructural roles are primarily responsible for the systems that distribute data and maintain its integrity. They work alongside data analytics professionals and help support their work. Their main responsibility is to ensure the functionality of data systems and the compliance with local, state, and federal regulations involving data security and ethics.

Key attributes

- **What they do:** Manage data sources and the overall data infrastructure
- **How they do it:** Work with the tools and databases used to manage data within a business

Sample job titles

- Data engineer
- Technology engineer
- Data manager
- Data steward
- IT architect

Business intelligence

Data analytics and business intelligence share a lot of commonalities. Both fields have professionals that use data to create insights that inform decision-making. A major difference is that business intelligence is more focused on creating processes and information channels that transform relevant data. Business intelligence professionals create tables, reports, and dashboards that empower stakeholders, giving them access to the data they need to inform the entire decision-making process on a continual basis. These roles often serve as a complement to core data analytics/data science professionals.

Key attributes

- **What they do:** Perform predictive analysis that enables organizations to determine likely future trends

- **How they do it:** Create tables, reports and dashboards that empower their organization

Sample job titles

- BI architect
- BI analyst
- BI solution developer
- BI software engineer
- Data viz & BI analyst

Additional roles for data professionals

Product development teams

The professionals in these roles manage a portfolio of customer and stakeholder analytic projects and initiatives. They often manage the analytical strategy for the organization. In these roles, experience is most likely required, and responsibilities are larger and more global.

Key attributes

- **What they do:** Manage analytical strategy within a project team
- **How they do it:** They are less hands-on with data analysis, serving as the person a data scientist or analysts would report to

Sample job titles

- Product manager
- Product developer
- Product lead
- Digital product manager
- Customer product manager

C-suite

This classification of roles covers high-ranking executives within an organization. The 'C' in c-suite stands for chief. In general, there's a trend for the c-suite to build data-driven decision making into their processes. Individuals filling these roles within organizational leadership teams are expected to be familiar with data and analytics.

Key attributes

- **What they do:** Responsible for data and data professionals across an entire organization
- **How they do it:** They are decision makers found at the top end of a company's hierarchy

Examples of job titles

- Chief marketing officer
- Chief data officer
- Chief analytics officer
- Chief information officer
- Chief data scientist

Key takeaways

There are a wide variety of roles and responsibilities in the data professional career space. Later, you will learn how these roles work together and collaborate with groups of professionals with diverse expertise on data analysis projects. Having a general understanding of the roles and responsibilities of data professionals can help inform your job search. This information can serve as a reference that can help you understand the needs and expectations of companies as you navigate through job postings. Take inventory of any transferable skills you may already possess from educational or professional experience.

Where data makes a difference for the future

As you have been learning, the data analytics field is dynamic, spanning a variety of industries. For you and other data professionals searching for new

job prospects, there are great opportunities across a spectrum of career fields. As you start to think about your own future role in data analytics, it can help to investigate how data is being used within different industries. In this reading, you'll consider some examples of industries and how data analytics helps guide them. You'll also consider the future of data analytics and how the field is still evolving!

There are so many different industries taking advantage of data analytics in so many different ways. Here is just a sample of some of these industries and how they use data:

Industry	Overview	How data is used
App-driven business (sharing economy service)	Facilitates users acquiring, providing, or sharing access to goods and services, often through online or app-based communities	• Maintaining functioning mobile applications • Delivering customized content based on user history including discounts • Using machine learning models to send notifications at key times or even locations
Automotive	Includes industries associated with the production, wholesaling, retailing, and maintenance of motor vehicles	• Gaining greater control over their supply chains • Improving production line performance, and designing new and more efficient vehicles • Enhancing vehicle safety and new features
Cybersecurity	Protects networks, devices, and data from unauthorized access or criminal use and the practice of maintaining confidentiality, integrity, and availability of information	• Locating weak points within networks and systems using predictive analytics • Defending against security attacks • Detecting data breaches through logic, models, and data tools • Improving the ability to identify attacks and respond to them with Artificial Intelligence (AI)
Digital marketing	Assists in advertising and promotional efforts of	• Translating customer interaction into actionable

	companies using the internet and online technologies	business data • Predicting user behaviors to personalize content and offers • Identifying patterns and trends that guide innovations • Determining the return on investment (ROI) of marketing efforts
Energy	Includes companies that explore, produce, refine, market, store, and transport both renewable and non-renewable energy resources	• Analyzing real-time data from power systems and monitoring devices • Optimizing technologies, monitoring power grids, and predicting failures • Preventing accidents and malfunctions
Gaming	Hosts an estimated 2.7 billion gamers worldwide, facilitating the interaction of players across the globe	• Designing world-building and character creation systems • Monitoring character engagement and how the environment reacts to player input • Optimizing game-play by identifying potential new features or upgrades • Regulating in-game purchases and fraud detection systems • Personalizing marketing campaigns
Streaming media and entertainment	Provides access to live and recorded content on-demand, delivered via the internet to computers, smart devices, and mobile devices	• Analyzing and monitoring user interactions to better understand customer sentiment • Matching users with advertisers with real-time analytics • Guiding future content decisions • Personalizing marketing campaigns

Telecommunications	Primarily involves operating and providing access to facilities for the transmission of voice, data, text, sound, and video	• Assisting the deployment, optimization, and predictive maintenance of telecommunications networks • Optimizing pricing models • Targeting advertisement and incentive campaigns, as well as detecting fraudulent activity • Analyzing customer data to customize subscriber plans
Travel and tourism	Encompasses a variety of services from transportation, accommodations, attractions, booking, and much more	• Marketing to individuals based on their previous travel or searched destinations • Directing machine learning systems that can adjust a traveler's itinerary based on set factors including weather and availability • Generating recommendations based on personal preferences and location-based discounts • Managing reservations and processing transactions

Data trends for the future

As you can already tell, data analytics is an emerging field with a wide range of exciting opportunities. And, even more exciting is the fact that big data is getting bigger. The need for people to understand, prioritize, manage, and analyze that information is not slowing down in any industry. Businesses will continue to rely on data-driven decision-making, fueled by both simple trend analyses and more complex techniques like predictive modeling and forecasting.

Additionally, more companies are storing all of their raw data within large repositories accessible across the organization. As companies become more reliant on insights generated by this data, there will be many opportunities for data analytics professionals to use their skills and knowledge to organize that information and make it useful.

Innovative technologies

Innovations in accessing this data are leading to new approaches in making data interconnected—meaning that there are still new and evolving ways businesses in different industries are going to use data in the future.

Technology is also ever-changing and adapting to these new needs. Because of this, there will always be exciting new tools and data solutions to explore.

Artificial intelligence will continue to have a large impact on business, helping to streamline many areas. For example, an increase in sales is understood to be a direct result of forecasting product demand. Artificial intelligence helps companies ensure warehouse supply, keep items in stock, reduce delivery time, and boost operational efficiency through automating processes.

Additionally, artificial intelligence will combine with machine learning, business intelligence, and automation to deliver more personalized services to customers.

Offering additional services will push forward innovation, bringing computer applications and the sources of stored data closer together physically. This concept is referred to as **edge computing**. By closing the gap between data and computation, speed improves. This results in greater support of real-time analytics and the automation necessary to support the increasing number of devices that are becoming linked through the internet of things.

An increasing number of data analytics tasks will be automated by creating, managing, and analyzing data in edge environments. Artificial intelligence and machine learning systems are only as equitable and inclusive as the people who create and train these systems. You will learn more about how you, as a data analytics professional, will need to take steps to ensure equity in the future.

Key takeaways

As you progress through your career as a data analytics professional, you will need to stay up-to-date with the latest trends and technologies used across different industries. The one certainty about the future of data analytics is that an increasing amount of data will continue to be generated and that new systems and innovations will continue to be developed, allowing data professionals an opportunity to learn, grow, and develop new skills.

Leverage data analysis in nonprofits

Nonprofit

- A group whose main purpose is to further a social cause or provide a benefit to the public

Open data

- Data that is available to the public and free to use, with guidance on how to navigate the datasets and acknowledge the source

Transportation

Ways data is collected

- Traffic sensors and cameras
- GPS and mapping apps
- Ride sharing apps

How data professionals use the data

- Create predictive analytics models
- Analyze the best way to get from one point to the next
- Determine the impact of development projects

Healthcare

Ways data is collected

- Electronic health records
- Personal health tracking devices
- Clinical trial data

How data professionals use the data

- Analyze medical imaging
- Predict genetic factors
- Create machine learning experiments to speed development of treatments

Finance

Ways data is collected

- Real-time analytics of spending activity
- Apps for banking and investing
- Electronic payments

How data professionals use the data

- Identify fraudulent behavior
- Facilitate payments
- Provide risk management solutions

Retail

Ways data is collected

- Sales data (online or in store)
- Retailer apps
- Customer loyalty programs

How data professionals use the data

- Recommend products
- Optimize inventory and pricing
- Market to individual customers

Restaurants

Ways data is collected

- Refrigerator temperature monitoring
- Reservation apps for customers
- Marketing/promotion responses

How data professionals use the data

- Monitor ingredients during storage
- Manage inventory and supply chain
- Anticipate staffing needs
- Gain feedback from customers

Utilities(Tiện ích công cộng)

Ways data is collected

- Sensors in pipeline/equipment deliver real-time data
- Drone data
- Interactive meters

How data professionals use the data

- Access usage data by both utility and customer
- Improve detection of risks within physical systems
- Reduce service interruptions